CSS3122 DATA STRUCTURE AND ALGORITHMS

CH5 ABSTRACT DATA STRUCTURE (PART 2)

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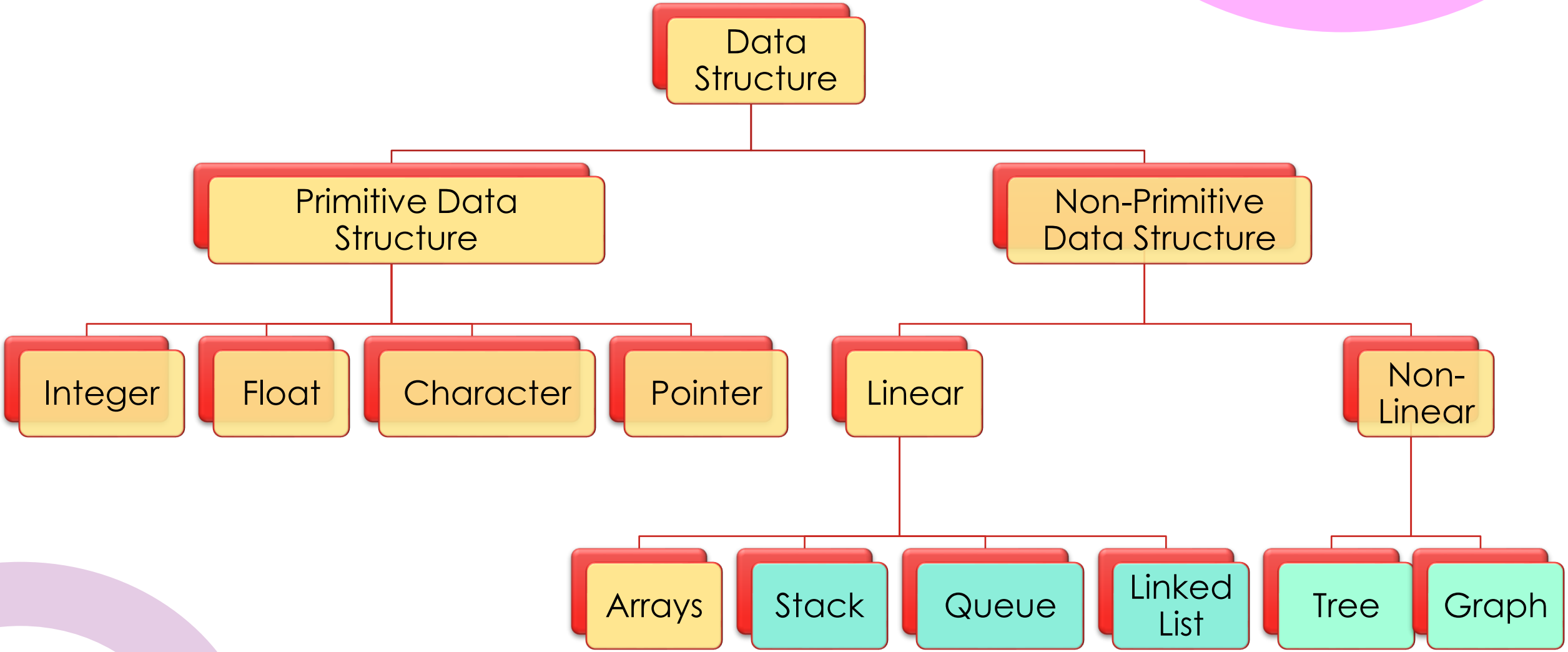
CHAPTER OUTLINE

In this chapter, you will learn about:

1. Linked List
2. Stack
3. Queue
4. Tree
5. Graph

INTRODUCTION



QUEUE

- Is a FIFO data structure – element that was inserted first will be the first to be taken out.
- Elements are added at one end called REAR.
- And removed from another end called FRONT.
- Can be implemented using array or linked list.

12	9	7	18	14	36				
0	1	2	3	4	5	6	7	8	9

A queue, with FRONT=0, REAR=5

QUEUE (CONT…)

12	9	7	18	14	36				
0	1	2	3	4	5	6	7	8	9

Original Queue

FRONT=0, REAR=5

12	9	7	18	14	36	45			
0	1	2	3	4	5	6	7	8	9

Queue after insertion of a new element

FRONT=0, REAR=6

	9	7	18	14	36	45			
0	1	2	3	4	5	6	7	8	9

Queue after deletion of an element

FRONT=1, REAR=6

QUEUE

- Before inserting an element into a queue:
  - Check for OVERFLOW condition.
  - i.e. queue is full.
  - i.e. when REAR = MAX – 1, where MAX is the queue size.

- Before deleting an element into a queue:
  - Check for UNDERFLOW condition.
  - i.e. queue is already empty.
  - i.e. when FRONT = -1 and REAR = -1.

QUEUE

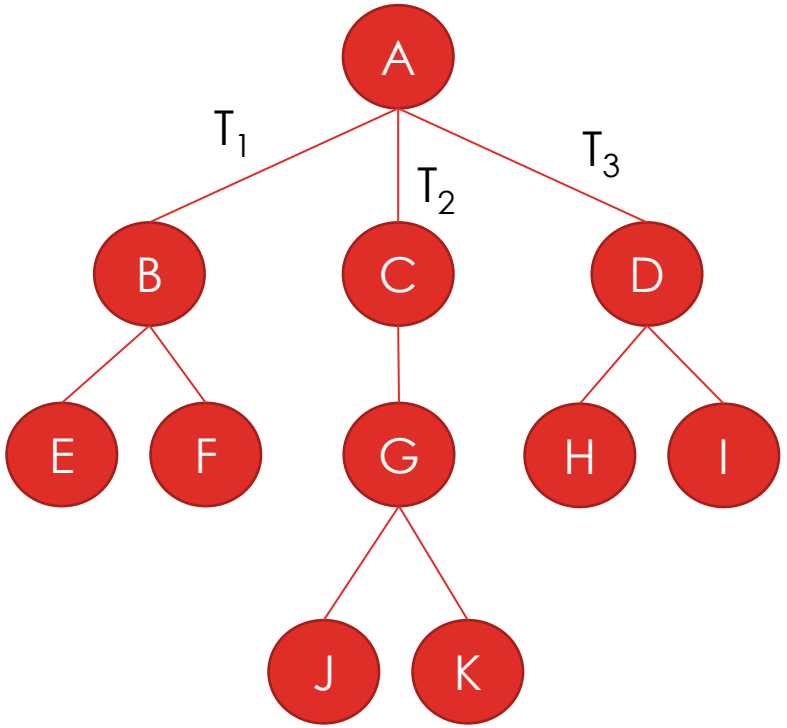
Algorithm to insert an element in a queue	
Step 1:	START
Step 2:	IF REAR = MAX-1 Write OVERFLOW Goto step 5 [END IF]
Step 3:	IF FRONT=-1 and REAR=-1 SET FRONT = REAR = 0 ELSE SET REAR = REAR + 1 [END OF IF]
Step 4:	SET QUEUE[REAR] = NUM
Step 5:	EXIT

Algorithm to delete an element from a queue	
Step 1:	START
Step 2:	IF FRONT=-1 or FRONT > REAR Write UNDERFLOW ELSE SET VAL=QUEUE[FRONT] SET FRONT=FRONT+1 [END IF]
Step 3:	EXIT

TREE

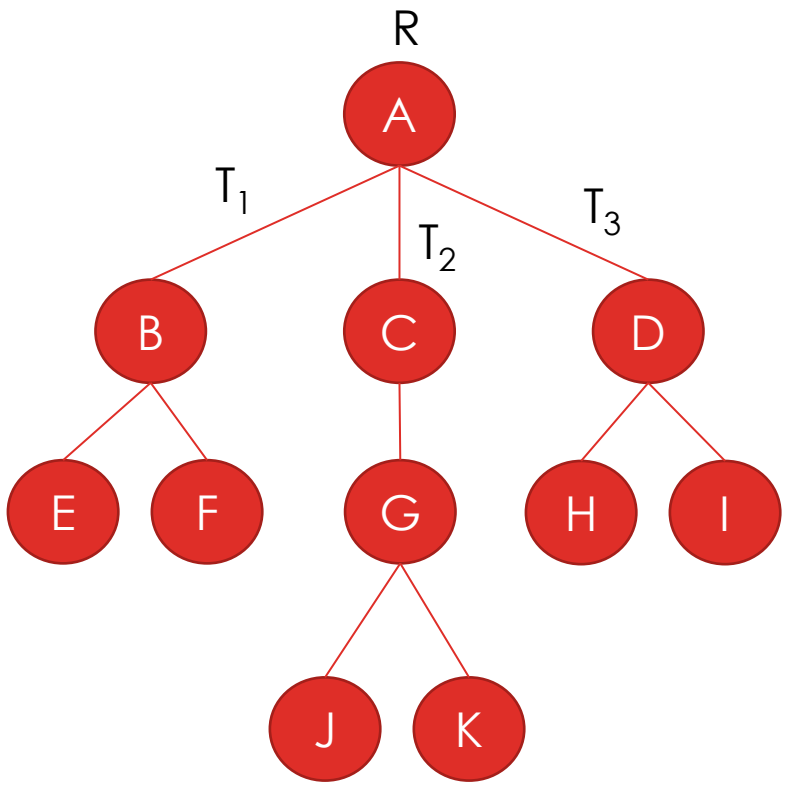
TREE

- Is a data structure that is a set of one or more nodes where one node is designated as the root of the tree and all the remaining nodes can be partitioned into non-empty sets of each which is a sub-tree of the root.



TREE (CONT…)

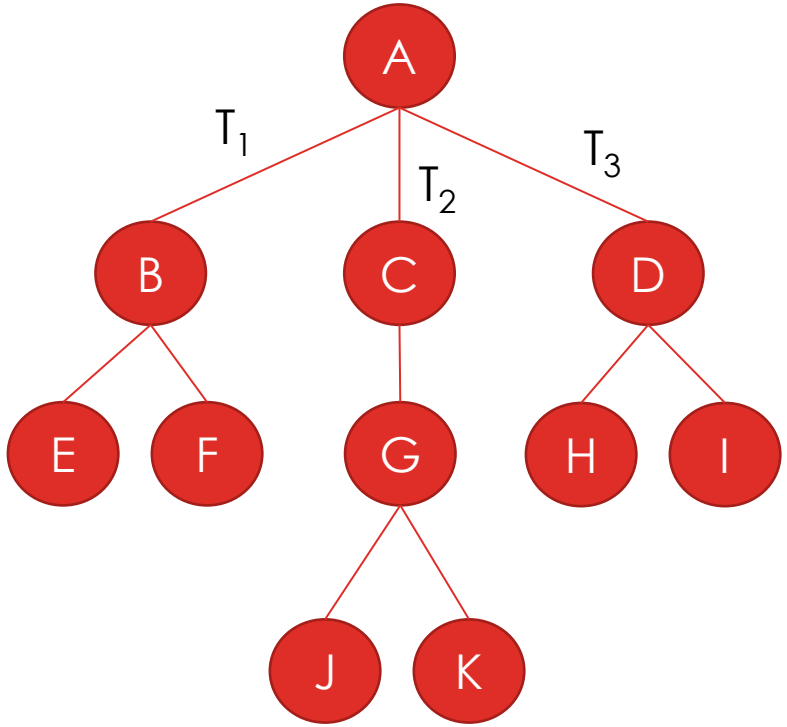
TREE TERMINOLOGY



- **Root node (R)**
  - topmost node in the tree.
  - If NULL, the tree is empty.
- **Sub-trees**
  - If R is not NULL, then trees T1, T2 and T3 are called sub-trees of R.
- **Leaf node**
  - Node that has no children.
  - Also called terminal node.
- **Path**
  - A sequence of consecutive edges.
  - E.g. the path from the root node A to node I is A, D, I.

TREE (CONT…)

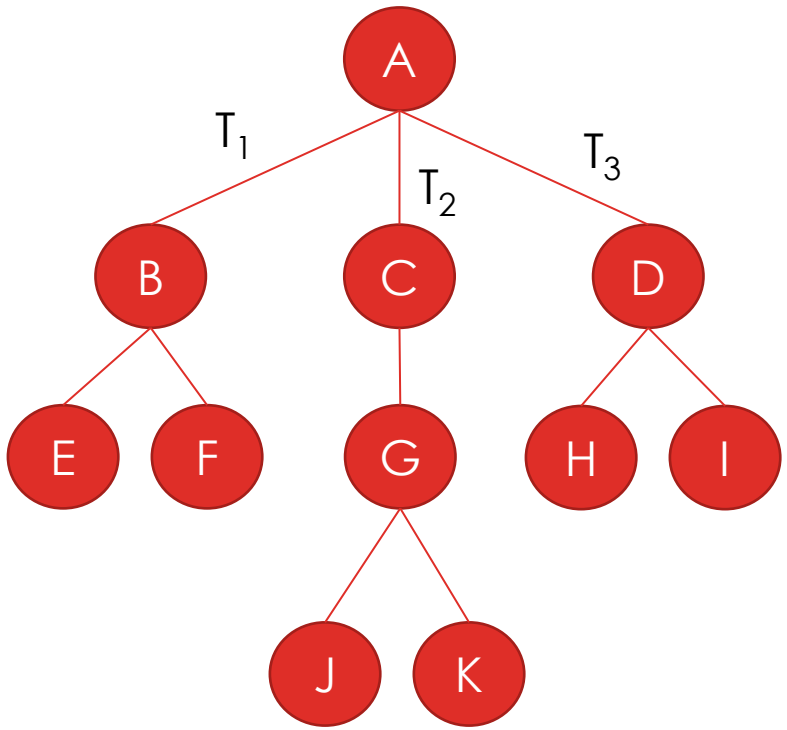
TREE TERMINOLOGY



- **Ancestor node**
  - Ancestor of a node is any predecessor node on the path from root to that node.
  - E.g. A, C and G are ancestor of K.
  - Root node does not have ancestors.
- **Descendant node**
  - A descendant is any successor node on any path from the node to a leaf node.
  - Leaf nodes do not have any descendants.
  - E.g. C, G, J and K are descendants of A.
- **Level number**
  - Every node in a tree is assigned a level number.
  - E.g. R is at level 0, children of R is at level 1.
- **Degree**
  - Equal to the number of children that a node has.
  - E.g. degree of G is 2, degree of a leaf node is 0.

TREE (CONT…)

TREE TERMINOLOGY



- **Siblings**
  - All nodes that are on the same level and share the same parent are called siblings (brothers).
  - E.g. H and I are siblings, J and K are siblings.
- **Edge**
  - The line connecting a node to its successors.
- **Depth**
  - The length of the path from the root R to the node.
  - E.g. the depth of R is 0.

TREE (CONT…)

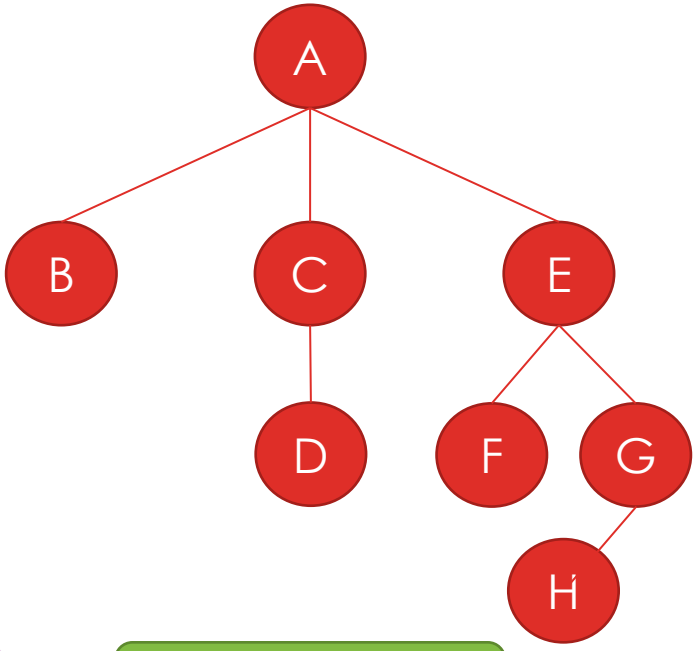
Types of trees

- Forests
- Binary trees
- Binary search trees (will be explained in upcoming chapter)
- Expression trees
- Tournament trees

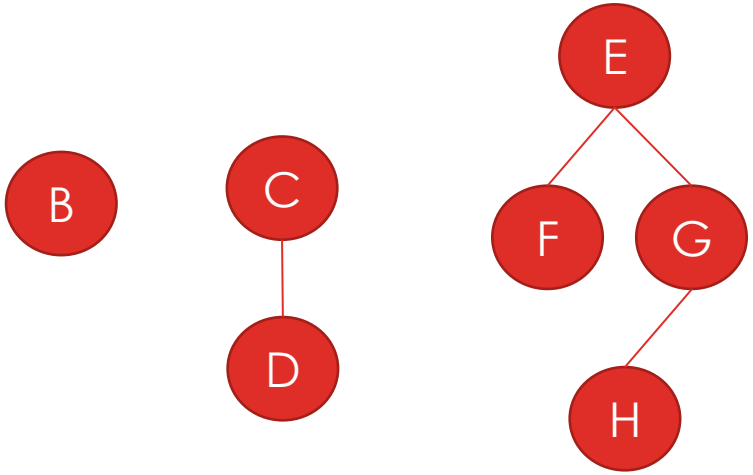
TREE (CONT…)

Forests

- Is a disjoint union of trees.
- Is obtained by deleting the root and the edges connecting the root node to nodes at level 1.



A TREE

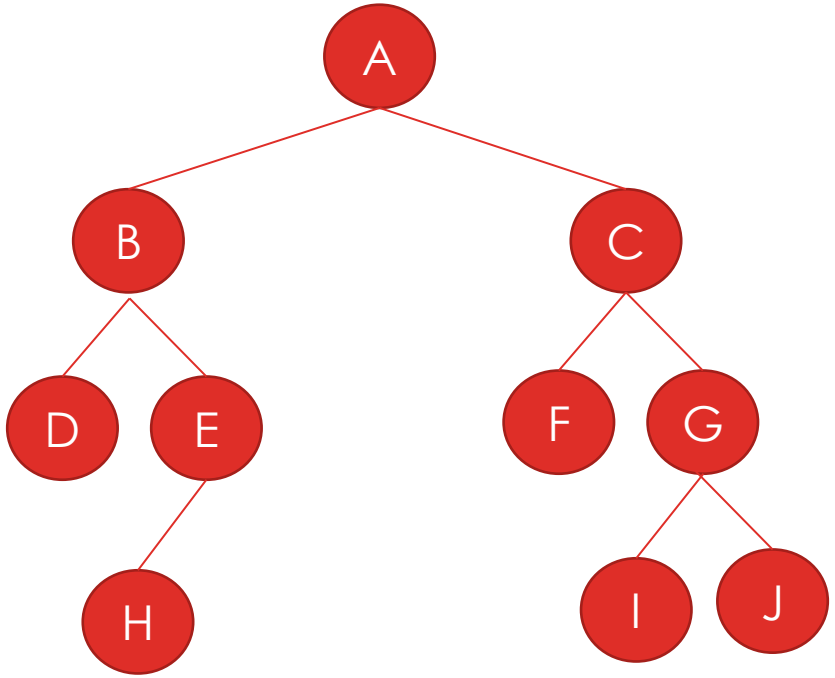


ITS CORRESPONDING FOREST

TREE (CONT…)

Binary Trees

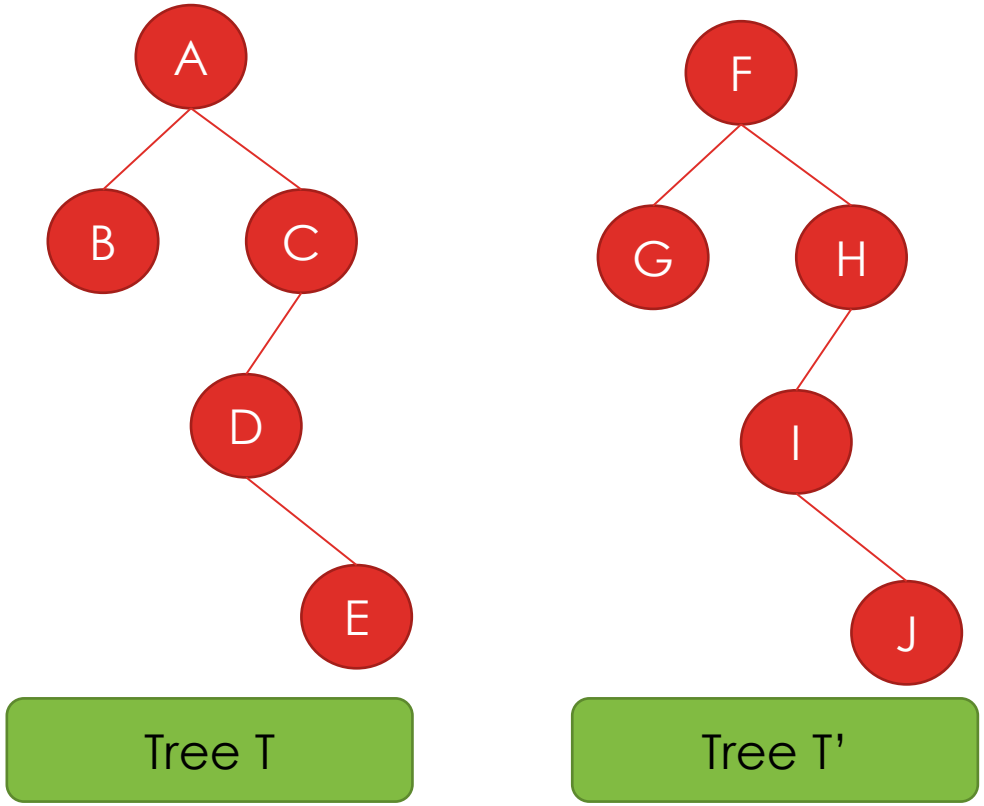
- A tree in which each node has either 0, 1 or at most 2 children.
- The root node is pointed by a ‘root’ pointer.
- Every node contains data elements, i.e. ‘left’ pointer which point to the left child, and ‘right’ pointer which point to the right child.
- For example:



TREE (CONT…)

Binary Trees (cont…)

- Similar binary trees
  - if both trees have the same structure.
  - E.g.

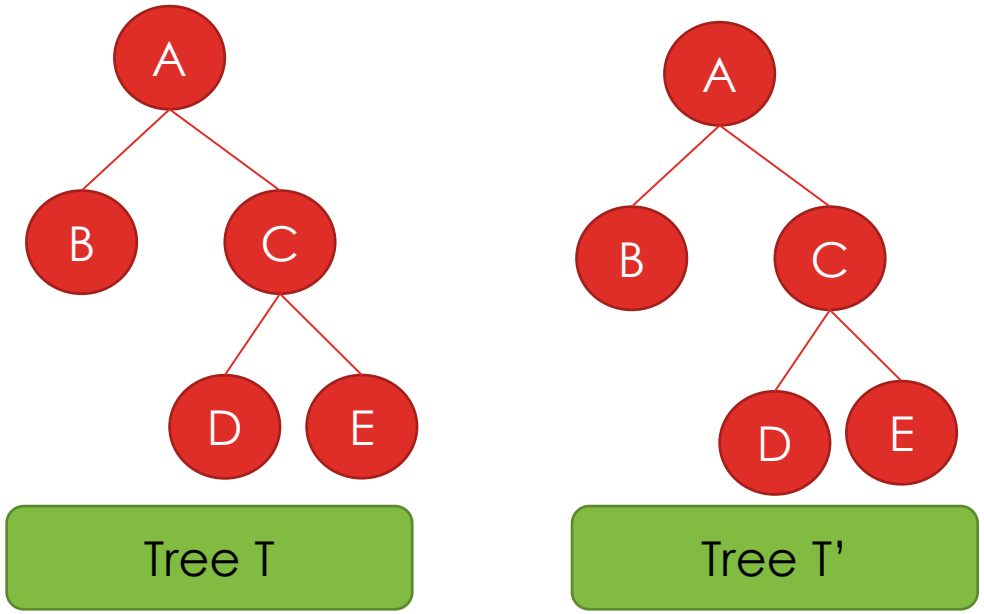


Tree T' is similar to T

TREE (CONT…)

Binary Trees (cont…)

- Copies of binary trees
  - if both trees have the similar structure and have same content at the corresponding nodes.
  - E.g.

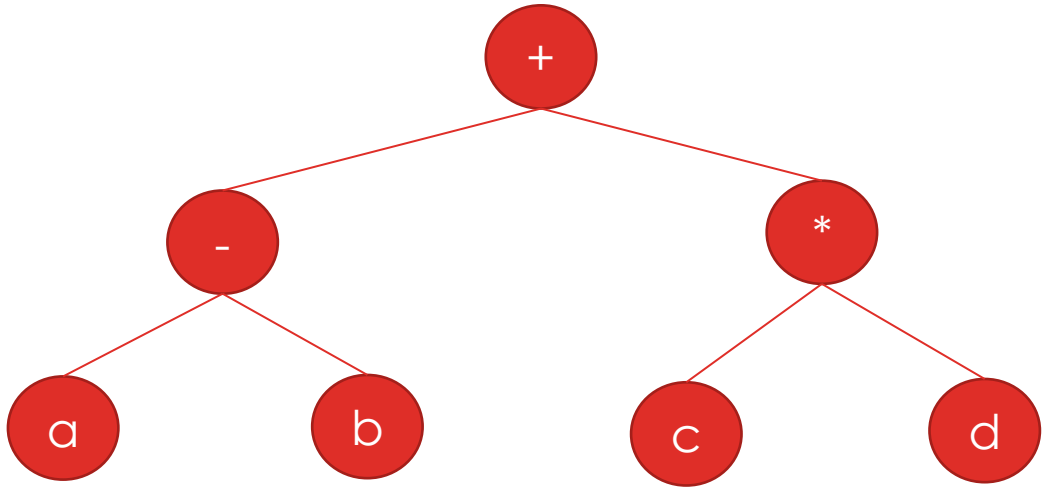


Tree T' is a copy of T

TREE (CONT…)

Espression Trees

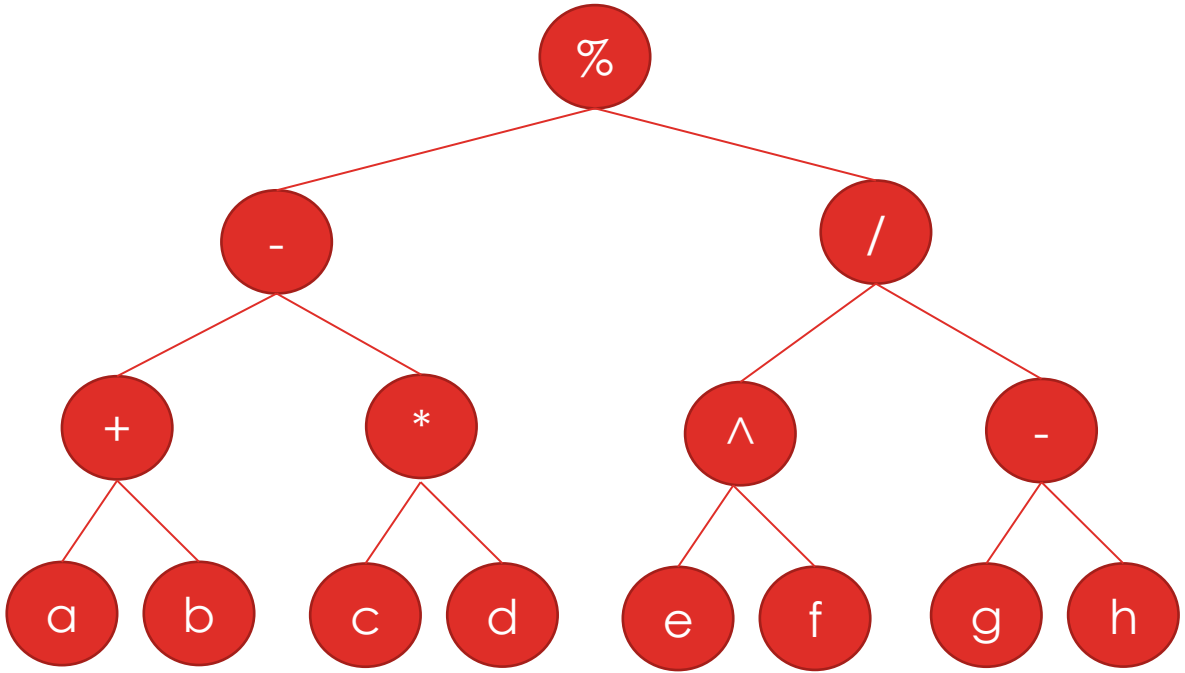
- Binary trees are widely used to store algebraic expressions.
- **EXAMPLE 1:**
- Given an expression as follows:
  - Exp=(a-b)+(c*d)
- The expression can be represented using a binary tree as follows:



TREE (CONT…)

Espression Trees (cont…)

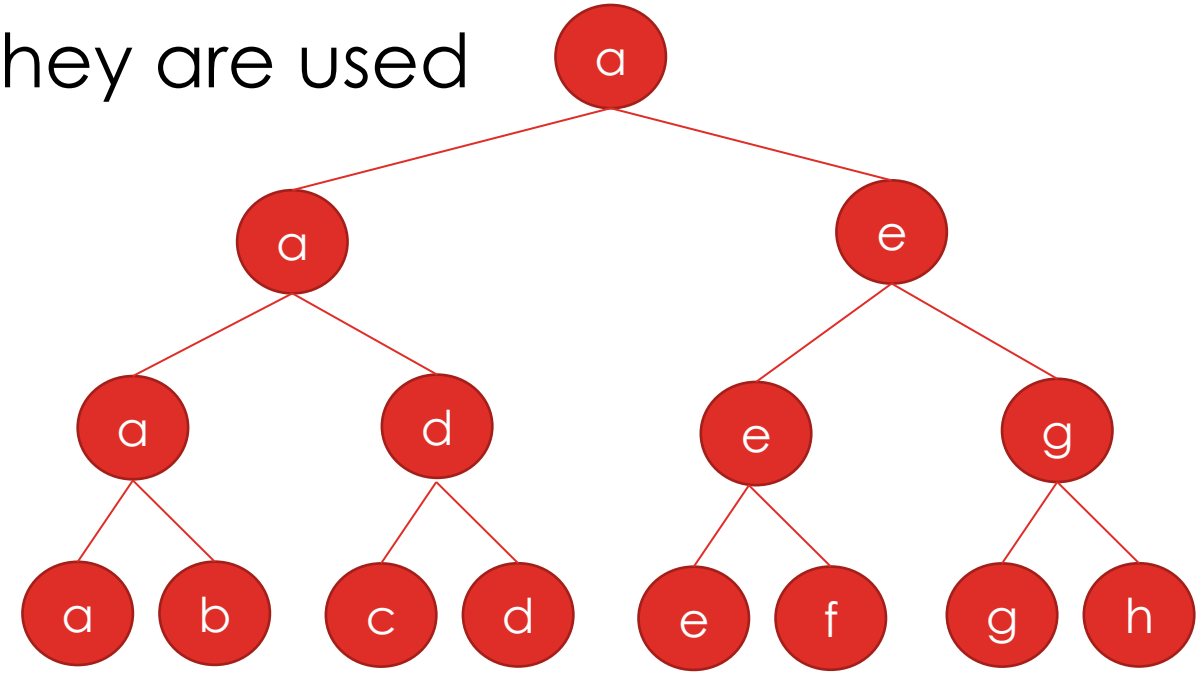
- **EXAMPLE 2:**
- Given an expression as follows:
  - Exp=((a+b)-(c*d))%((e^f)/(g-h))
- Construct the corresponding binary tree:



TREE (CONT…)

Tournament Trees

- Also called as selection trees.
- Each external node represents a player and each internal nodes represents the winner of the match played between the players represented by its children nodes.
- Also called as 'winner' trees as they are used to record winner at each level.



- In the tree, there are 8 players; **a, b, c, d, e, f, g & h**.
- In round 1, a compete against b, c against d, e against f, and g against h.
- And **a is the final winner** of the matches.

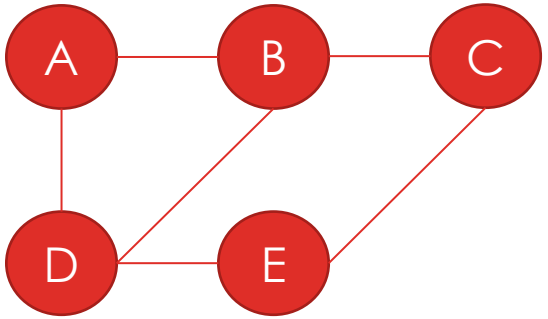
GRAPH

- Is an ADT that is used to implement mathematical concept of graphs.
- Is basically a collection of nodes and edges that connect the nodes.
- Graph is often seen as a generalization of the tree structure, where instead of having a parent-to-child relationship, any kind of complex relationship can exist.
- E.g. graph can be used to depict 'transportation networks' – in which nodes are airports, intersections, ports, etc. The edges can be airline flights, one-way roads, shipping routes, etc.

GRAPH (CONT…)

DEFINITION

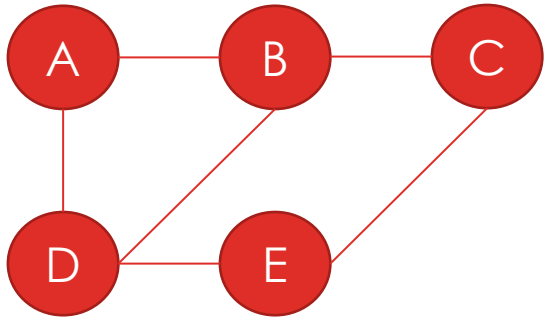
- A graph G is defined as an ordered set (V, E), where V(G) represents the set of vertices (nodes) and E(G) represents the edges that connect these vertices.
- Example:



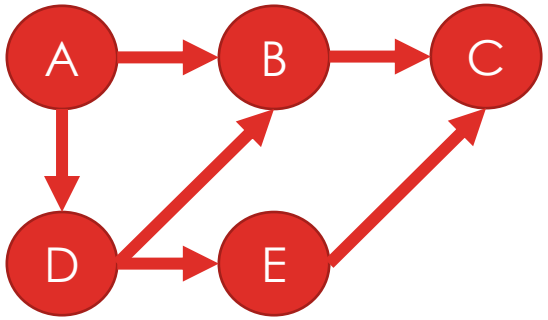
- There are 5 vertices and 6 edges in the graph.
- A graph with V(G) = {A, B, C, D, E} and E(V,G)={(A,B),(B,C),(A,D),(B,D),(D,E),(C,E)}.

GRAPH (CONT…)

- A graph G can be directed or undirected.

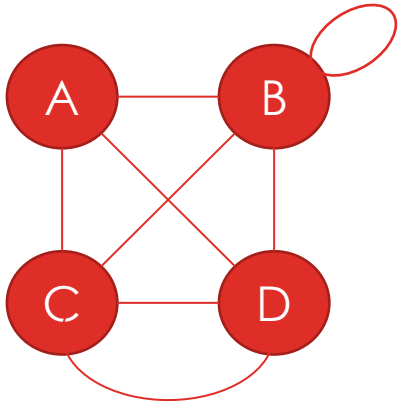


Undirected



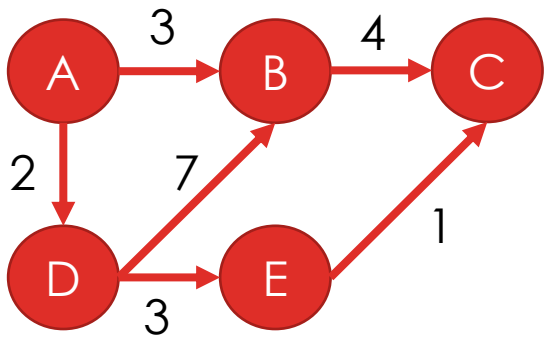
Directed

- Multi-graph - a graph with multiple edges and/or loops.



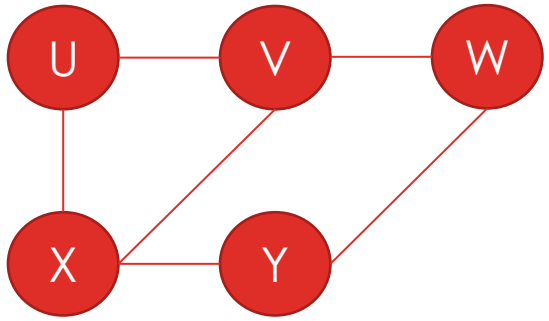
GRAPH (CONT…)

- Labelled or weighted graph – if every edge in the graph is assigned some data.



GRAPH (CONT…)

GRAPH TERMINOLOGY:

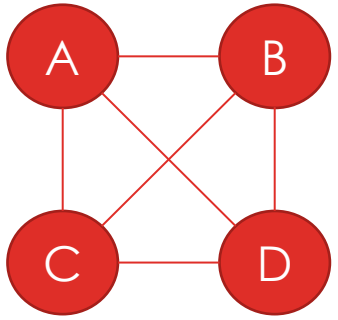


Undirected

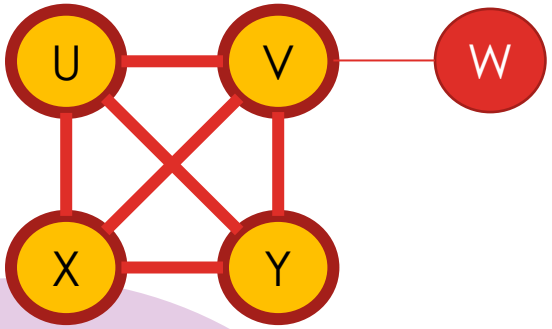
- **Adjacent nodes or neighbours**
  - 2 nodes with an edge that connects them together.
  - The 2 nodes are said to be the end terminals of the edge.
  - E.g. the nodes U and V are said to be adjacent/neighbours.
- **Degree of a node**
  - The number of edges containing the node.
  - E.g. degree of U is 2.

GRAPH (CONT…)

GRAPH TERMINOLOGY:



A complete graph

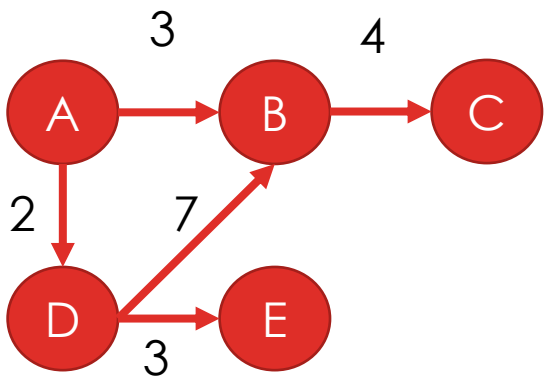


A clique in a graph

- **Complete graph**
  - When all of its nodes are fully connected.
  - i.e. there is a path from one node to every other node in the graph.
- **Clique**
  - A subset of the vertices where for every 2 vertices in the graph, there is an edge that connects them.

GRAPH (CONT…)

GRAPH TERMINOLOGY:

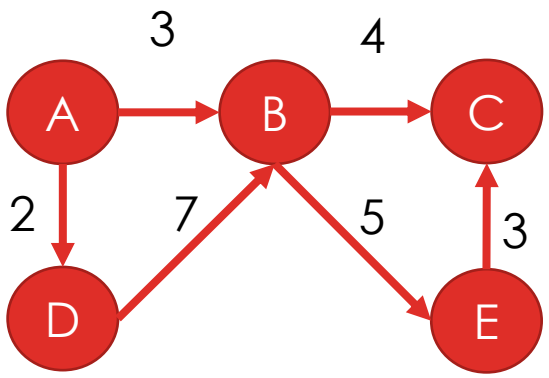


A directed graph

- **Out-degree of a node**
  - No of edges that originates at a node.
  - E.g. out-degree of D is 2.
- **In-degree of a node**
  - No of edges that terminate at a node.
  - E.g. in-degree of D is 1.
- **Pendant vertex**
  - A vertex with degree 1.
  - E.g. E and C are pendant vertices.
- **Isolated vertex**
  - A vertex with degree 0.

GRAPH (CONT…)

GRAPH TERMINOLOGY:



A directed graph

- **Cut vertex**
  - A node, where if deleted from the graph will disconnect the graph.
  - E.g. B is a cut vertex.
- **Source**
  - If a node has a +ve out-degree but 0 in-degree.
  - E.g. A is a source node.
- **Sink**
  - If a node has a +ve in-degree but 0 out-degree.
  - E.g. C is a sink node.

GRAPH (CONT…)

Graph traversal algorithms:

- Traversal of a graph is a method that is used to examine the nodes and edges of a graph.
- In this chapter, you will learn about these 2 algorithms:
  1. Breadth-first search
  2. Depth-first search